

Project Report

Bachelor of Computer Applications (BCA)

Semester – IV

Sub: Data Warehousing and Mining

Topic: Online Retail Data Analysis using Power BI

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ONLINE RETAIL DATA ANALYSIS USING POWER BI

Abstract

This project presents a comprehensive data warehouse and analytics solution for an online retail business using Microsoft Power BI. The dataset, sourced from Kaggle, contains transactional records of an online retail store spanning multiple years and countries. The primary objective is to extract meaningful business insights from raw retail data through effective data preprocessing, data modeling, DAX calculations, and interactive dashboard development.

The project demonstrates the application of ten key DAX functions — SUM, AVERAGE, COUNT, DISTINCTCOUNT, IF, CALCULATE, ALL, FILTER, MAX, and MIN — to build analytical measures. Five interactive dashboards were developed covering overview metrics, trend analysis, category performance, geographical distribution, and detailed reporting.

Introduction

In today's data-driven business environment, organizations generate vast amounts of transactional data that, when analyzed effectively, can provide significant competitive advantages. This project focuses on analyzing an online retail dataset using Microsoft Power BI. The dataset is the well-known Online Retail dataset available on Kaggle, containing approximately 3,92,692 rows of transactional data from a UK-based online retailer.

Problem Statement

Online retail businesses generate large volumes of transactional data daily. Without proper analysis, this data remains underutilized. Key challenges include lack of visibility into revenue performance, inability to identify top-performing products, no clear understanding of domestic vs international revenue, and limited geographical insights.

- How to clean and prepare raw retail transaction data for analysis
- How to identify key revenue drivers and sales trends
- How to analyze customer purchasing behavior
- How to visualize business performance effectively using dashboards

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Dataset Description

The dataset is sourced from Kaggle and contains transactional data from a UK-based online retail store.

Dataset Characteristics:

- Time Period: December 2010 – December 2011
- Records: ~500,000 transactions
- Nature: Real-world e-commerce data

Total Records: 3,92,692 rows

Time Period: 2010–2011

Key Columns: Invoice_ID, Customer_ID, Country, Region, Revenue, Unit_Price, Qty_Sold, Product_Name, Is_International, Revenue_Tier, Order_Year, Month Name

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Data Preprocessing

Removed duplicates and null values

Filtered negative quantities and prices

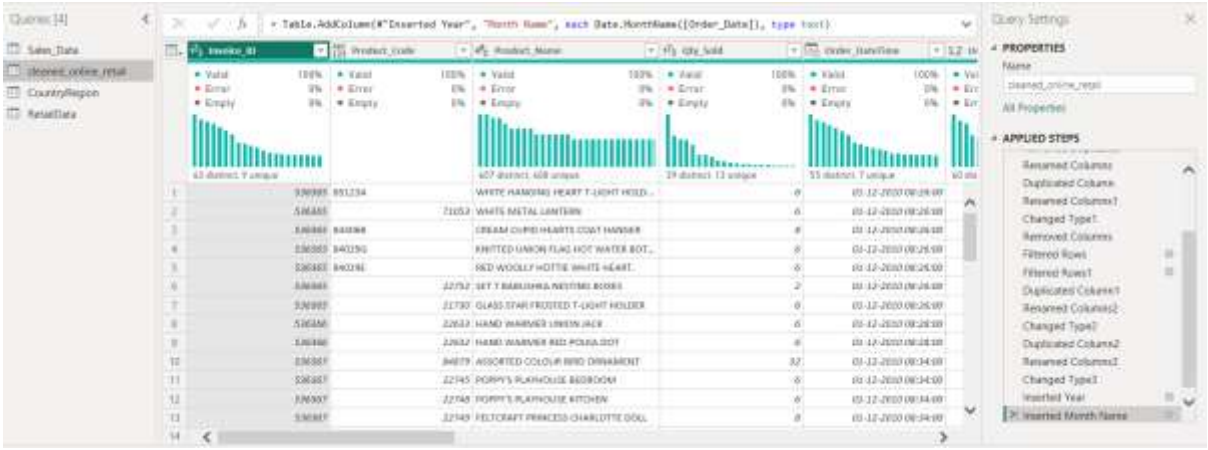
Created Revenue column (Qty_Sold × Unit_Price)

Added Month Name, Order_Year, Year columns

Classified Revenue_Tier (Low Value / High Value)

Added Is_International and Order_Size columns

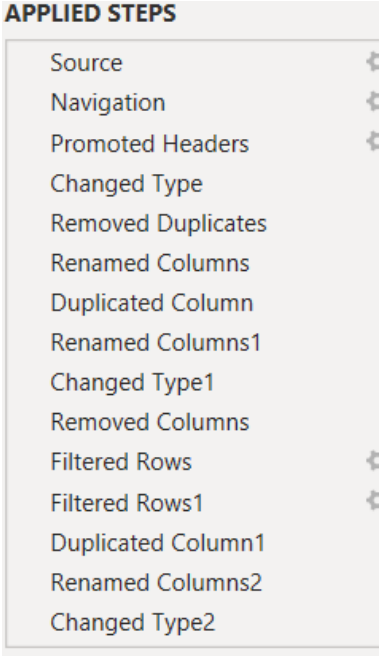
Standardized country and region names



Data Modeling

A **star schema model** was implemented for efficient analysis.Star schema design with cleaned_online_retail as the fact table, CountryRegion as dimension table. Relationships defined on Country column for cross-filtering between tables.

This structure improves query performance and simplifies dashboard creation.



DAX Calculations

DAX (Data Analysis Expressions) was used to derive meaningful metrics.

Total Revenue = SUM(cleaned_online_retail[Revenue])

Avg Unit Price = AVERAGE(cleaned_online_retail[Unit_Price])

Total Orders = COUNT(cleaned_online_retail[Invoice_ID])

Unique Customers = DISTINCTCOUNT(cleaned_online_retail[Customer_ID])

Revenue Performance = IF([Total Revenue] >= 50000, "Above Target", "Below Target")

Domestic Revenue = CALCULATE(SUM(...Revenue), Is_International = FALSE())

% of Total Revenue = DIVIDE(SUM(...Revenue), CALCULATE(SUM(...Revenue), ALL(...)))

International Revenue = CALCULATE(SUM(...Revenue), FILTER(..., Is_International = TRUE()))

Max Qty Sold = MAX(cleaned_online_retail[Qty_Sold])

Min Unit Price = MIN(cleaned_online_retail[Unit_Price])

Total Revenue	Total Orders	Unique Customers	Avg Unit Price	Domestic Revenue	International Revenue	Max Qty Sold	Min Unit Price
88,87,208.89	392692	4338	3.13	72,85,024.64	16,02,184.25	80995	0.00

Dashboard Development

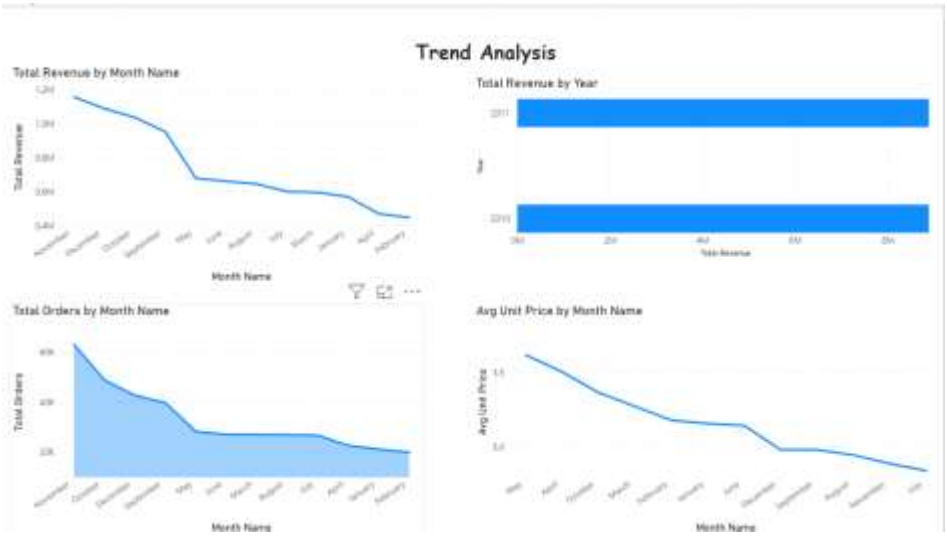
Overview Dashboard

The Overview Dashboard provides a high-level summary of the overall business performance. It displays key KPI cards showing Total Revenue, Total Orders, Unique Customers, and Average Unit Price. A Bar Chart visualizes revenue distribution across countries, while a Donut Chart highlights the split between Domestic and International revenue. This dashboard gives stakeholders a quick snapshot of how the business is performing at a glance.



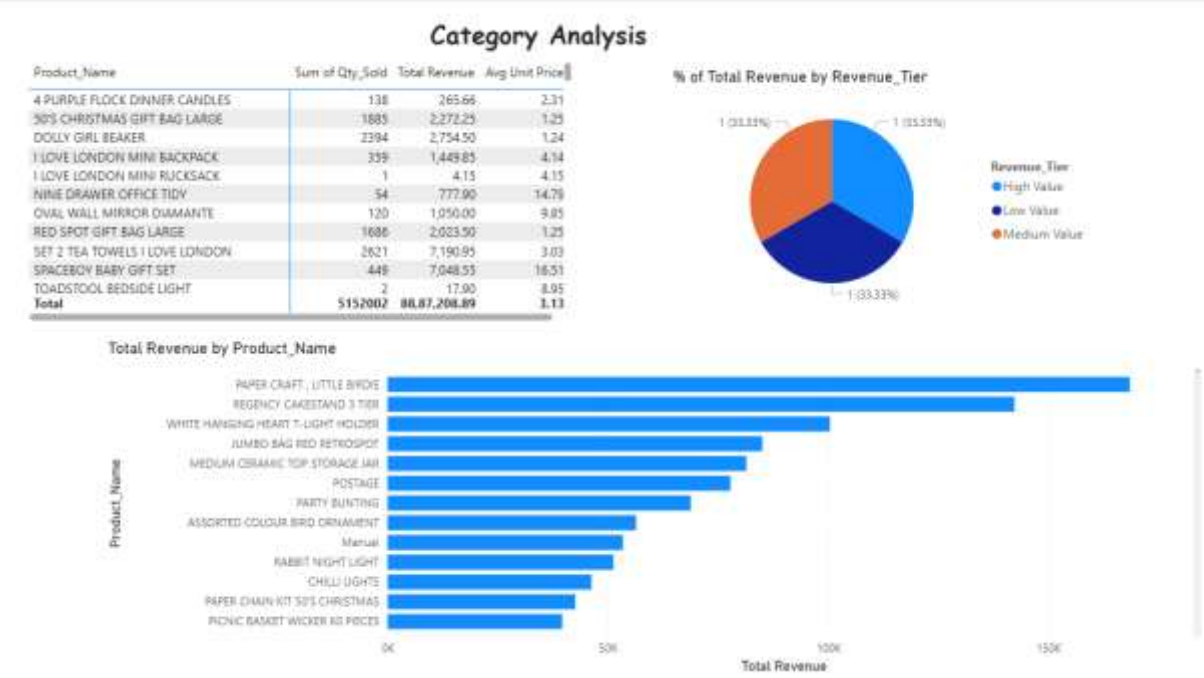
Trend Analysis

The Trend Analysis Dashboard focuses on understanding how revenue and orders change over time. A Line Chart displays monthly revenue trends, an Area Chart shows order volume patterns across months, and a Bar Chart compares total revenue across different years. This dashboard helps identify seasonal peaks, growth patterns, and year-over-year performance to support forecasting and planning decisions.



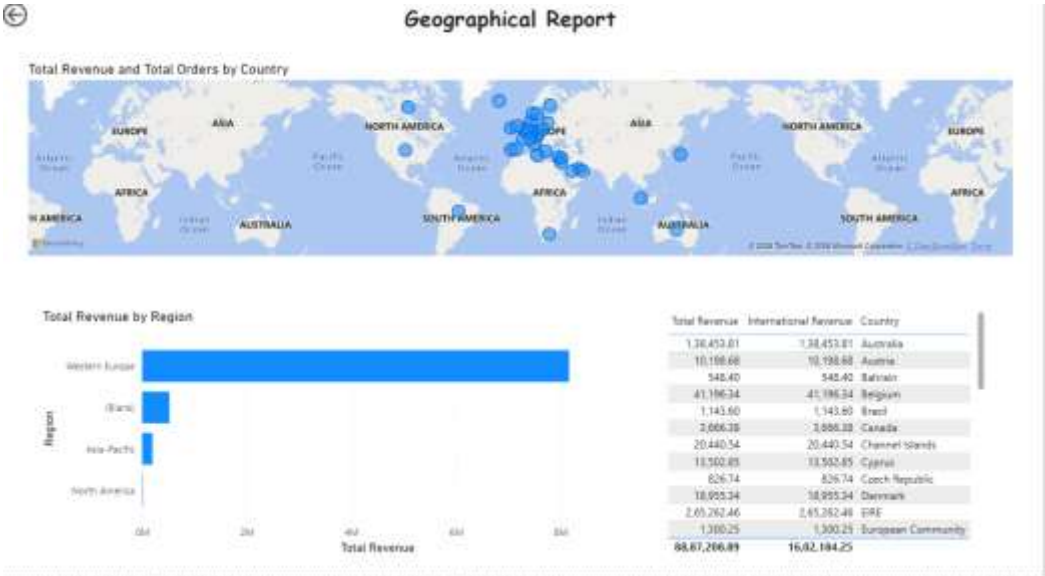
Category Analysis

The Category Analysis Dashboard breaks down performance by product and revenue tier. A Bar Chart ranks products by total revenue, a Pie Chart shows the percentage contribution of each Revenue Tier (Low Value and High Value), and a Matrix Table provides a detailed product-level view of revenue, average unit price, and maximum quantity sold. This dashboard helps identify top-performing products and underperforming categories.



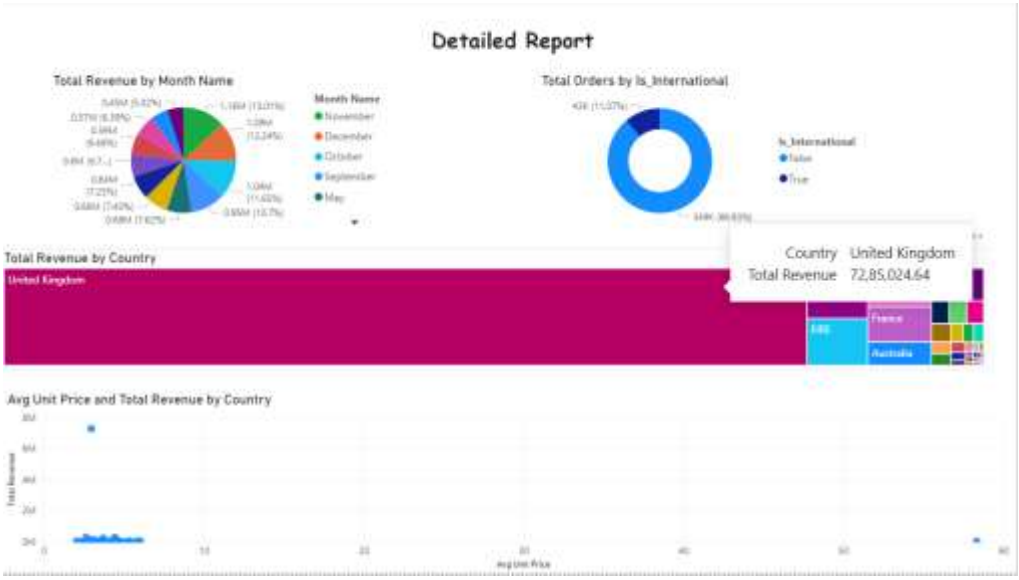
Geographical Report

The Geographical Report Dashboard visualizes revenue distribution across countries and regions worldwide. A Map Visual displays revenue bubbles across the world map, a Bar Chart compares total revenue by region, and a Table breaks down Domestic Revenue vs International Revenue by country. This dashboard helps identify key markets, expansion opportunities, and regional performance gaps.



Detailed Report

The Detailed Report Dashboard provides a granular drill-down view of the data at product and customer level. A Matrix Table allows drill-down from Country to Product level, while multiple Donut and Pie Charts show revenue split by country, revenue tier, international vs domestic orders, and monthly revenue distribution. This dashboard is designed for in-depth analysis and supports detailed business reporting requirements.



Benefits

- ✓ Western Europe drives majority of revenue
- ✓ Seasonal purchasing patterns identified
- ✓ Domestic orders outweigh international orders
- ✓ High-value products identified for prioritization
- ✓ Faster decision-making through interactive dashboards

Challenges Faced

- Handling missing CustomerID values
- Managing large dataset size (~500k rows)
- Identifying and filtering canceled transactions
- Maintaining data consistency during cleaning
- Designing an efficient data model

Conclusion

This project demonstrates the power of data analytics in transforming raw data into actionable insights. Using Microsoft Power BI, the dataset was successfully cleaned, modeled, and visualized.

The insights derived from this analysis can help businesses optimize their strategies, improve customer satisfaction, and increase profitability. The project highlights the importance of data-driven decision-making in modern business environments.

References

Kaggle Online Retail Dataset:
https://www.kaggle.com/datasets/ulrikthygepedersen/online-retail-dataset

Microsoft Power BI Docs: https://docs.microsoft.com/en-us/power-bi/

DAX Reference: https://docs.microsoft.com/en-us/dax/